2. Data

2.1 Data Description

The data for this project represents a comprehensive and diverse collection of data sources related to credit applications and repayment behaviors, making it both rich in detail and complex to work with. This data includes current loan applications with static demographic and financial information, Credit Bureau reports with historical data from other financial institutions, monthly balances on various credit accounts such as credit cards and POS (point of sale) loans, and detailed repayment histories that capture installment payments, including missed or late payments, across multiple credit types. Additionally, it includes information on previous loan applications within Home Credit, providing an in-depth look at each client's financial profile and credit history over time. The combination of these sources results in a broad dataset that offers valuable insights into repayment patterns and client behavior but also presents significant challenges.

A primary challenge with this data is its heterogeneity; each source captures unique aspects of a client's financial behavior, often in different formats. Some datasets provide static information, while others track monthly balances, leading to a mix of snapshot and time-series data that must be carefully aligned. Additionally, the data spans inconsistent collection periods, complicating synchronization across timeframes. This complexity requires thoughtful feature engineering to capture each client's creditworthiness effectively. Below, we describe each of the tables in detail.

Application_train.csv and ***application_test.csv***

(307k rows, 122 columns) is the primary data set for this project. This file contains data about clients and their current application and information about the applicant. There is a similar 'application_test.csv' file containing our test data (49k rows, 121 columns, it does not contain TARGET). After dropping some redundant/highly correlated columns, the following columns might be retained from this data:

Variable	**Description**
AMT_ANNUITY	Loan annuity
AMT_CREDIT	Credit amount of the loan
AMT_INCOME_TOTAL	Income of the client

Variable	Description
AMT_REQ_CREDIT_BUREAU_DAY	Number of enquiries to Credit Bureau about the client one day before application (excluding one hour before application)
AMT_REQ_CREDIT_BUREAU_HOUR	Number of enquiries to Credit Bureau about the client one hour before application
AMT_REQ_CREDIT_BUREAU_MON	Number of enquiries to Credit Bureau about the client one month before application (excluding one week before application)
AMT_REQ_CREDIT_BUREAU_QRT	Number of enquiries to Credit Bureau about the client 3 months before application (excluding one month before application)
AMT_REQ_CREDIT_BUREAU_WEEK	Number of enquiries to Credit Bureau about the client one week before application (excluding one day before application)
AMT_REQ_CREDIT_BUREAU_YEAR	Number of enquiries to Credit Bureau about the client one year (excluding last 3 months before application)
APARTMENTS_AVG	Normalized information about building where the client lives. What is average (_AVG suffix), modus (_MODE suffix), median (_MEDI suffix) apartment size, common area, living area, age of building, number of elevators, number of entrances, state of the building, number of floors
BASEMENTAREA_AVG	Normalized information about building where the client lives. What is average (_AVG suffix), modus (_MODE suffix), median (_MEDI suffix) apartment size, common area, living area, age of building, number of elevators, number of entrances, state of the building, number of floors

Variable	**Description**
CNT_CHILDREN	Number of children the client has
CODE_GENDER	Gender of the client
COMMONAREA_AVG	Normalized information about building where the client lives. What is average (_AVG suffix), modus (_MODE suffix), median (_MEDI suffix) apartment size, common area, living area, age of building, number of elevators, number of entrances, state of the building, number of floors
DAYS_BIRTH	Client's age in days at the time of application
DAYS_EMPLOYED	How many days before the application the person started current employment
DAYS_ID_PUBLISH	How many days before the application did client change the identity document with which they applied for the loan
DAYS_LAST_PHONE_CHANGE	How many days before application did client change phone
DAYS_REGISTRATION	How many days before the application did client change their registration
DEF_30_CNT_SOCIAL_CIRCLE	How many observations of client's social surroundings defaulted on 30 DPD (days past due)
ENTRANCES_AVG	Normalized information about building where the client lives. What is average (_AVG suffix), modus (_MODE suffix), median (_MEDI suffix) apartment size, common area, living area, age of building, number of elevators, number of entrances, state of the building, number of floors

Variable	**Description**
EXT_SOURCE_1 - 3	Normalized score from external data source
FLAG_CONT_MOBILE	Was mobile phone reachable (1=YES, 0=NO)
FLAG_DOCUMENT_1 - 21	Did client provide document 1 - 21
FLAG_EMAIL	Did client provide email (1=YES, 0=NO)
FLAG_OWN_CAR	Flag if the client owns a car
FLAG_OWN_REALTY	Flag if client owns a house or flat
FLAG_PHONE	Did client provide home phone (1=YES, 0=NO)
FLAG_WORK_PHONE	Did client provide work phone (1=YES, 0=NO)
FLOORSMAX_AVG	Normalized information about building where the client lives. What is average (_AVG suffix), modus (_MODE suffix), median (_MEDI suffix) apartment size, common area, living area, age of building, number of elevators, number of entrances, state of the building, number of floors
FLOORSMIN_AVG	Normalized information about building where the client lives. What is average (_AVG suffix), modus (_MODE suffix), median (_MEDI suffix) apartment size, common area, living area, age of building, number of elevators, number of entrances, state of the building, number of floors
LANDAREA_AVG	Normalized information about building where the client lives. What is average (_AVG suffix), modus (_MODE suffix), median (_MEDI suffix) apartment size, common area, living area, age of building,

Variable	Description
	number of elevators, number of entrances, state of the building, number of floors
LIVE_CITY_NOT_WORK_CITY	Flag if client's contact address does not match work address (1=different, 0=same, at city level)
NAME_CONTRACT_TYPE	Identification if loan is cash or revolving
NAME_EDUCATION_TYPE	Level of highest education the client achieved
NAME_FAMILY_STATUS	Family status of the client
NAME_HOUSING_TYPE	What is the housing situation of the client (renting, living with parents, etc.)
NAME_INCOME_TYPE	Client's income type (businessman, working, maternity leave, etc.)
NAME_TYPE_SUITE	Who was accompanying client when they applied for the loan
NONLIVINGAPARTMENTS_AVG	Normalized information about building where the client lives. What is average (_AVG suffix), modus (_MODE suffix), median (_MEDI suffix) apartment size, common area, living area, age of building, number of elevators, number of entrances, state of the building, number of floors
NONLIVINGAREA_AVG	Normalized information about building where the client lives. What is average (_AVG suffix), modus (_MODE suffix), median (_MEDI suffix) apartment size, common area, living area, age of building, number of elevators, number of entrances, state of the building, number of floors

Variable	Description
OBS_30_CNT_SOCIAL_CIRCLE	How many observations of client's social surroundings with observable 30 DPD (days past due) default
OCCUPATION_TYPE	What kind of occupation the client has
ORGANIZATION_TYPE	Type of organization where client works
OWN_CAR_AGE	Age of client's car
REG_REGION_NOT_WORK_REGION	Flag if client's permanent address does not match work address (1=different, 0=same, at region level)
REGION_POPULATION_RELATIVE	Normalized population of region where client lives (higher number means the client lives in a more populated region)
REGION_RATING_CLIENT	Rating of the region where client lives (1,2,3)
SK_CURR_ID	ID of loan in our sample
TARGET	Target variable (1 - client with payment difficulties: they had a late payment more than X days on at least one of the first Y installments of the loan in our sample, 0 - all other cases)
YEARS_BEGINEXPLUATATION_AVG	Normalized information about building where the client lives. What is average (_AVG suffix), modus (_MODE suffix), median (_MEDI suffix) apartment size, common area, living area, age of building, number of elevators, number of entrances, state of the building, number of floors
YEARS_BUILD_AVG	Normalized information about building where the client lives. What is average (_AVG suffix), modus (_MODE suffix), median (_MEDI suffix) apartment

Variable	Description
	size, common area, living area, age of building, number of elevators, number of entrances, state of the building, number of floors

Previous_application.csv

(1.7 million rows, 37 columns) This table contains information about clients' previous credit applications for Home Credit loans of clients who have loans in the train and test data sets. There is one row for each previous application.One SK_ID_CURR can have multiple rows of previous applications in this table. After dropping some redundant/highly correlated columns, and aggregating the following columns might be retained from this data:

Variable	Description
SK_ID_CURR	ID of loan in our sample
PREV_AMT_APPLICATION_MEAN	Mean of how much credit the client asked for on previous applications
PREV_AMT_APPLICATION_SUM	Total amount of how much credit the client asked for on all previous applications
PREV_AMT_CREDIT_MEAN	Mean of how much credit the client was approved for on previous applications
PREV_AMT_CREDIT_SUM	Total amount of how much credit the client was approved for on previous applications
PREV_AMT_DOWN_PAYMENT_MEAN	Mean of down payments on all previous applications
PREV_AMT_DOWN_PAYMENT_SUM	Total amount of down payments on all previous applications
PREV_AMT_ANNUITY_MEAN	Mean of annuity for all previous applications
PREV_AMT_ANNUITY_SUM	Total sum of annuity for all previous applications

Variable	Description
PREV_CNT_PAYMENT_MEAN	Mean of terms of previous credit at application of the previous applications
PREV_CNT_PAYMENT_SUM	Total sum of terms of previous credit at application of the previous applications
PREV_APPROVED_SUM	Total number of approved applications for SK_ID_CURR
PREV_REFUSED_SUM	Total number of refused applications for SK_ID_CURR
PREV_CANCELED_SUM	Total number of canceled applications for SK_ID_CURR
PREV_APPROVAL_RATE	Total number of approved applications / Total number of applications
PREV_CREDIT_APPLICATION_RATIO_MEAN	PREV_AMT_CREDIT_MEAN / PREV_AMT_APPLICATION_MEAN

credit_card_balance.csv

(3.8 million rows, 23 columns) This dataset contains monthly data about previous credit cards clients have had with Home Credit. Each row is one month of a credit card balance, and a single credit card can have many rows. After dropping some redundant/highly correlated columns, the following columns might be retained from this data:

Variable	Description
SK_ID_PREV	ID of previous credit in Home Credit related to loan. (One loan in their sample can have 0, 1, 2, or more previous loans in Home Credit)
SK_ID_CURR	ID of loan in their sample

Variable	Description
MONTHS_BALANCE	Month of balance relative to application date (-1 means the freshest balance date)
AMT_BALANCE	Balance during the month of previous credit
AMT_CREDIT_LIMIT_ACTUAL	Credit card limit during the month of the previous credit
AMT_DRAWINGS_CURRENT	Amount drawing during the month of the previous credit
AMT_PAYMENT_TOTAL_CURRENT	How much did the client pay during the month in total on the previous credit
CNT_DRAWINGS_CURRENT	Number of drawings during this month on the previous credit
CNT_INSTALMENT_MATURE_CUM	Number of paid installments on the previous credit
NAME_CONTRACT_STATUS	Contract status (Active, Approved, Completed, Demand, Refused, Sent proposal, Signed) on the previous credit
SK_DPD_DEF	DPD (Days past due) during the month with tolerance (debts with low loan amounts are ignored) of the previous credit

Bureau.csv

(1.7 million rows, 17 columns) This table contains information about clients' previous credits provided by other financial institutions. One SK_ID_CURR can have multiple rows of previous credit in this table. After dropping some redundant/highly correlated columns, the following columns might be retained from this data:

Variable	Description
SK_ID_CURR	ID of loan in the sample

Variable	Description
SK_BUREAU_ID	ID used to join with bureau_balance features if needed
CREDIT_ACTIVE	Status of Credit Bureau reported credits
DAYS_CREDIT	Number of days before the current application when the client applied for Credit Bureau credit
CREDIT_DAY_OVERDUE	Number of days past due on Credit Bureau credit at the time of application for the related loan in the sample
DAYS_CREDIT_ENDDATE	Remaining duration of CB credit (in days) at the time of application
AMT_CREDIT_MAX_OVERDUE	Maximum amount overdue on the Credit Bureau credit at the application date of the loan in the sample
CNT_CREDIT_PROLONG	Number of times the Credit Bureau credit was prolonged
AMT_CREDIT_SUM	Current credit amount for the Credit Bureau credit
AMT_CREDIT_SUM_DEBT	Current debt on the Credit Bureau credit
AMT_CREDIT_SUM_LIMIT	Current credit limit of a credit card reported in Credit Bureau
AMT_CREDIT_SUM_OVERDUE	Current amount overdue on the Credit Bureau credit
CREDIT_TYPE	Type of credit (e.g., car loan, cash loan, etc.)
AMT_ANNUITY	Annuity of the Credit Bureau credit

Bureau_balance.csv

(27 million rows, 3 columns) This table contains monthly balances for previous credits. There is one row for each month of history for every previous credit that was reported to the credit bureau.

Variable	Description
SK_BUREAU_ID	ID used to join with bureau features if needed
MONTHS_BALANCE	Month of balance relative to application date
STATUS	Status of the credit loan during the month

POS_CASH_balance.csv

(10 million rows, 8 columns) This dataset provides monthly balance snapshots for previous Point of Sale (POS) and cash loans that the applicants had with Home Credit. Each row represents one month of repayment history for a given credit, including both consumer credit and cash loans linked to loans in the current sample. The dataset captures detailed, time-based information on each credit account's status, payment terms, and any overdue payments. Each entry is associated with a unique credit and represents one monthly observation, covering the loan's status, remaining installments, and potential overdue days.

Variable	Description
SK_ID_PREV	Identifier for each previous credit associated with the loans in the sample. This allows linking multiple rows to a single previous credit, recording monthly snapshots of the credit status.
SK_ID_CURR	Identifier for the current loan in the sample, connecting each record back to the specific loan application.
MONTHS_BALANCE	The number of months relative to the loan application date for each balance snapshot. A value of -1 represents the most recent monthly snapshot. This field allows for chronological tracking of credit history.
CNT_INSTALMENT	The number of installments in the original term of the previous credit.
CNT_INSTALMENT_FUTURE	The number of remaining installments on the previous credit. This variable helps track repayment progress and indicates how much of the loan is still unpaid.

Variable	Description
NAME_CONTRACT_STATUS	The status of the credit contract during each monthly snapshot. Possible statuses include “Active,” “Completed,” and “Past Due,” among others. This variable is useful for tracking whether the loan was repaid on time, still ongoing, or completed.
SK_DPD	Days Past Due (DPD) during the month for the previous credit. This field indicates the number of days by which the payment was overdue in that month, if any, providing insight into the client’s payment discipline.
SK_DPD_DEF	Days Past Due with a tolerance level (small overdue amounts are ignored) for the previous credit during the month. This metric helps identify significant overdue periods while ignoring minor payment delays that may not indicate real risk.

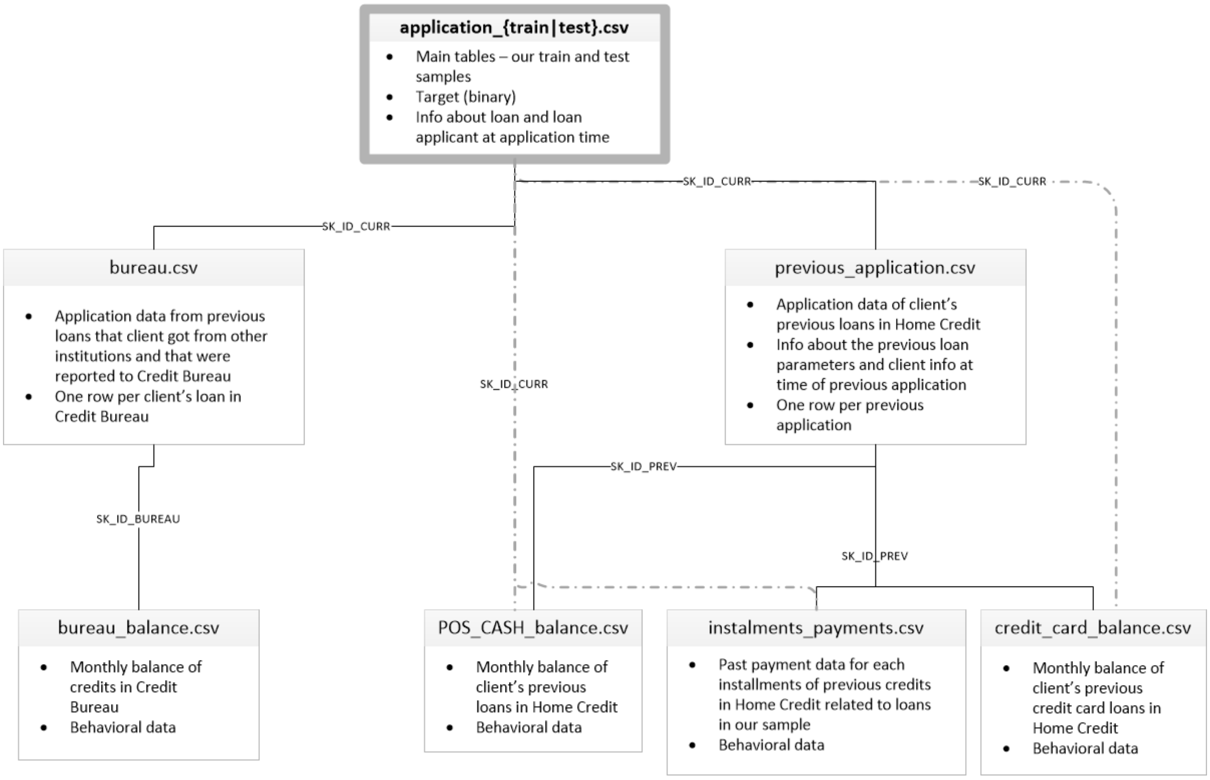
Installments_payments.csv

(13.6 million rows, 8 columns) The installments_payments.csv file contains detailed repayment history data for previously disbursed credits related to the loans in the sample. Each row corresponds to a single installment, capturing both payments made by clients and instances where payments were missed. Each entry represents a record of one payment for an installment or a record of a missed installment for a previous Home Credit credit related to the loans in our sample.

Variable	Description
SK_ID_PREV	An identifier for each previous credit related to the loans in the sample. Multiple rows may be associated with a single credit, reflecting each installment or payment related to that credit.
SK_ID_CURR	A unique identifier for the current loan in the sample. This links each installment record back to the specific loan application.

Variable	Description
NUM_INSTALMENT_VERSION	Indicates the version of the installment calendar. A value of 0 denotes a credit card-related installment.
NUM_INSTALMENT_NUMBER	The specific installment number within the credit's payment schedule. This indicates the position of the installment within the entire repayment plan.
DAYS_INSTALMENT	The scheduled day for the installment payment, relative to the application date of the current loan. A negative value represents days before the loan application date.
DAYS_ENTRY_PAYMENT	The actual day the payment was made, relative to the application date of the current loan. This allows for comparisons between the scheduled payment date and the actual payment date to assess payment timeliness.
AMT_INSTALMENT	The prescribed (expected) payment amount for this particular installment. This reflects the amount due according to the credit agreement.
AMT_PAYMENT	The actual payment amount made by the client for the installment. Comparing AMT_PAYMENT with AMT_INSTALMENT provides insight into whether clients overpaid, underpaid, or missed payments.

Data Diagram


application_{train|test}.csv
Main tables – our train and test samples
Target (binary)
Info about loan and loan applicant at application time
SK_ID_CURR
SK_ID_CURR
SK_ID_CURR
SK_ID_CURR
bureau.csv
Application data from previous loans that client got from other institutions and that were reported to Credit Bureau
One row per client's loan in Credit Bureau
previous_application.csv
Application data of client's previous loans in Home Credit
Info about the previous loan parameters and client info at time of previous application
One row per previous application
SK_ID_BUREAU
SK_ID_PREV
SK_ID_PREV
bureau_balance.csv
Monthly balance of credits in Credit Bureau
Behavioral data
POS_CASH_balance.csv
Monthly balance of client's previous loans in Home Credit
Behavioral data
instalments_payments.csv
Past payment data for each installments of previous credits in Home Credit related to loans in our sample
Behavioral data
credit_card_balance.csv
Monthly balance of client's previous credit card loans in Home Credit
Behavioral data